

Software plane

consumes a native device interface

native driver $\xrightarrow{\text{observes}}$ $\mathcal{C}_e$ logical contract

Control plane

defines and mediates the interface

$\mathcal{T} \longrightarrow \beta_e \longrightarrow \mu_e$
software-chosen topology, binding, and semantic adaptation

virtualization

host software back-end
+ FPGA front-end

Data plane

executes and transfers payload

$M_{\text{DUT}} \xleftrightarrow{\text{DMA payload}} P_e$ physical execution

selective passthrough

SDN analogy: programmable control over a separately realized data path; not an SDN application API.